

PRACTICAL COURSE WORKBOOK

HTML & CSS Foundations

Build and test a responsive digital product

LEVEL	Beginner
GUIDED TIME	6 hours across 12 lessons
PROJECT	a responsive, accessible application you can demonstrate
SKILLS	HTML, CSS, Flexbox, Grid, Responsive design, Accessibility testing
EDITION	Structured draft - version 0.9

WHAT YOU WILL BE ABLE TO DO

- Create valid semantic documents.
- Build accessible forms and content.
- Use the cascade, box model, Flexbox and Grid deliberately.
- Test responsive behavior, accessibility and performance.
- Publish a portfolio with reproducible release evidence.

WHO THIS IS FOR

Beginners who can manage files and use a browser, and who want a defensible first web project.

BEFORE YOU BEGIN

- Create and rename plain-text files
- Use a modern browser
- Install or access a text editor

Use this workbook with the online lessons. It is designed for A4 printing, handwritten evidence notes, and offline revision. It does not provide certification.

WORKBOOK GUIDE

A clear route through the course

Build and publish a responsive, accessible portfolio using standards-based HTML and CSS. The course moves from document structure through layout, accessibility, performance and release evidence.

MODULE	LESSONS AND FOCUS
01 HTML foundations	1. Build a valid HTML document 2. Structure content with semantic HTML 3. Write links, lists, images and media
02 Forms and CSS foundations	1. Create an accessible contact form 2. Control the CSS cascade 3. Size and space elements predictably
03 Modern layout	1. Lay out one dimension with Flexbox 2. Build two-dimensional layouts with Grid 3. Design responsive pages around content
04 Production quality	1. Test keyboard, focus, contrast and reflow 2. Improve images, fonts and rendering performance 3. Validate, test and publish the portfolio

HOW TO USE EACH LESSON PAGE

- 1 Read the objective and identify the decision it supports.
- 2 Attempt the retrieval prompt before checking the online explanation.
- 3 Reproduce the worked example with the stated input.
- 4 Test one normal case and one boundary or failure case.
- 5 Record the result, limitation and next action in the evidence log.

PRINT AND FILE

Print at 100% scale on A4 paper. Use double-sided printing with long-edge binding. Keep completed pages with the project files named in the evidence log.

MODULE 01

HTML foundations

Create meaningful, interoperable documents.

LESSON 01 / 30 MIN

Build a valid HTML document**Learning objectives**

- Explain the browser-facing purpose of the document skeleton.
- Create a valid page with language, character encoding, viewport and title metadata.
- Inspect parsing and metadata with developer tools.

Key terms

doctype, head, body

Retrieval prompt

Which combination gives a mobile browser and assistive technology essential document context?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 02 / 30 MIN

Structure content with semantic HTML**Learning objectives**

- Choose elements by meaning rather than appearance.
- Build a logical landmark and heading structure.
- Check the page outline and reading order.

Key terms

landmarks, headings, source order

Retrieval prompt

Why should a section normally have a heading?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 03 / 30 MIN

Write links, lists, images and media**Learning objectives**

- Write links whose purpose is clear out of context.
- Choose useful alternative text based on the image's role.
- Use lists and figures for the relationships they represent.

Key terms

link purpose, alternative text, list semantics

Retrieval prompt

What is the best alt value for a purely decorative divider image?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 02

Forms and CSS foundations

Build interaction and control styling and sizing.

LESSON 04 / 30 MIN

Create an accessible contact form

Learning objectives

- Associate every control with a persistent label.
- Group related choices with fieldset and legend.
- Use native input types and understandable validation.

Key terms

label, fieldset, validation

Retrieval prompt

Which attribute makes an input value available under a key during form submission?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 05 / 30 MIN

Control the CSS cascade

Learning objectives

- Predict which declaration wins.
- Use inheritance and custom properties deliberately.
- Avoid escalating selector specificity.

Key terms

cascade, specificity, inheritance

Retrieval prompt

When two same-origin, unlayered class rules have equal specificity, what normally decides the winner?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 06 / 30 MIN

Size and space elements predictably

Learning objectives

- Calculate the rendered box size.
- Apply consistent sizing and spacing.
- Diagnose overflow instead of hiding it.

Key terms

box model, box-sizing, overflow

Retrieval prompt

With box-sizing:border-box, what does width:300px include?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 03

Modern layout

Use Flexbox, Grid and responsive constraints.

LESSON 07 / 30 MIN

Lay out one dimension with Flexbox

Learning objectives

- Choose Flexbox for row or column distribution.
- Control growth, shrinking and alignment.
- Preserve usable source order and wrapping.

Key terms

main axis, cross axis, wrapping

Retrieval prompt

In the default flex-direction, which property distributes free horizontal space?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 08 / 30 MIN

Build two-dimensional layouts with Grid

Learning objectives

- Choose Grid for rows and columns.
- Create fluid tracks without device-specific breakpoints.
- Prevent content from forcing tracks wider than the viewport.

Key terms

grid tracks, minmax, auto-fit

Retrieval prompt

What does 1fr represent in a grid track?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 09 / 30 MIN

Design responsive pages around content

Learning objectives

- Start with a fluid, narrow-screen layout.
- Add breakpoints where content needs them.
- Verify reflow and zoom without horizontal scrolling.

Key terms

viewport, media query, reflow

Retrieval prompt

When should a new breakpoint normally be introduced?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 04

Production quality

Audit accessibility, optimize delivery and publish defensible work.

LESSON 10 / 30 MIN

Test keyboard, focus, contrast and reflow

Learning objectives

- Complete the interface using only a keyboard.
- Provide visible, unobscured focus.
- Measure contrast and test reflow.

Key terms

keyboard, focus indicator, contrast

Retrieval prompt

Why is an automated accessibility scan insufficient on its own?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 11 / 30 MIN

Improve images, fonts and rendering performance

Learning objectives

- Deliver appropriately sized images.
- Prevent avoidable layout movement.
- Measure before and after optimization.

Key terms

responsive images, layout shift, critical rendering path

Retrieval prompt

What is the purpose of the sizes attribute when srcset uses width descriptors?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 12 / 30 MIN

Validate, test and publish the portfolio

Learning objectives

- Run a repeatable pre-release checklist.
- Test normal, boundary and failure conditions.
- Publish an evidence-backed portfolio release.

Key terms

validation, cross-browser testing, deployment

Retrieval prompt

What is the strongest evidence that a release is ready?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

CAPSTONE PROJECT

Design, build, test and publish a professional one-page portfolio.

DELIVERABLE	A deployed portfolio, editable source, accessibility audit and versioned release record.
TOOLS	Text editor, Modern browsers and developer tools, Nu HTML Checker, Accessibility and contrast testing tools

Production plan

- 1 Define audience, content and success criteria.
- 2 Create semantic HTML and an accessible contact form.
- 3 Build fluid CSS with Flexbox and Grid.
- 4 Test keyboard, focus, zoom, reflow, contrast and reduced motion.
- 5 Optimize images and measure loading behavior.
- 6 Publish a preview and complete the release record.

Definition of done

- HTML validation has no unresolved errors.
- The complete journey works with keyboard alone.
- Content reflows at 320 CSS pixels and 400% zoom without loss.
- Text and component contrast meet the selected WCAG 2.2 target.
- Images have correct alternatives, dimensions and responsive candidates.
- Release evidence names environments, results and limitations.

PROJECT EVIDENCE

Document decisions another person can review

1. PURPOSE AND USER

State the intended user, need, expected outcome and constraints.

2. INPUTS AND ASSUMPTIONS

List source files, data, versions, permissions and assumptions.

PROJECT EVIDENCE

Tests, limitations and revision

3. NORMAL CASE

Expected result, observed result and evidence location.

4. BOUNDARY OR FAILURE CASE

What was difficult, what happened and why it matters.

5. REVISION AND LIMITATION

What changed, what improved and what remains unproven.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can a new visitor identify the owner, work and contact action?
- Does source order remain logical without CSS?
- Can every action be completed by keyboard?
- Which measured limitation should be fixed next?

FINAL DECISION

Is the project ready to share? State the evidence, remaining risk, owner and next review date.

Authoritative references

Learn web development

MDN Web Docs - reviewed 2026-09-04

https://developer.mozilla.org/en-US/docs/Learn_web_development

HTML Standard

WHATWG - reviewed 2026-09-04

<https://html.spec.whatwg.org/>

Page Structure Tutorial

W3C Web Accessibility Initiative - reviewed 2026-09-04

<https://www.w3.org/WAI/tutorials/page-structure/>

Web Content Accessibility Guidelines 2.2

W3C - reviewed 2026-09-04

<https://www.w3.org/TR/WCAG22/>

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Ask two representative users to complete the portfolio journey, record observations and revise the highest-impact barrier.